

Entelect

Graduate Programme Team

Dear Hiring Team,

I am writing to apply for Entelect's 2026 Graduate Programme. I am currently completing my Honours degree in Information Technology at North-West University and expect to graduate in November 2026. I am particularly interested in Entelect because of the Boot Camp's practical, hands-on approach to developing graduates. The opportunity to work on real client projects while learning from experienced software engineers is exactly how I want to start my career.

During my studies, I have developed a strong foundation in full-stack software development and machine learning. For my Honours research, I am developing a multimodal machine learning framework to support the early detection of foot-and-mouth disease outbreaks in South Africa. This project combines computer vision, natural language processing and environmental data, and has strengthened my skills in Python, PyTorch and OpenCV while teaching me how to solve complex, real-world problems.

I have also built full-stack web applications using PHP, MySQL and JavaScript. One of my key projects was a Campus Marketplace platform that integrated Paystack payments and Shiplogic courier services, giving me practical experience working with third-party APIs and building complete software solutions. In addition, I gained industry exposure as an IT Vacation Student at Iclix (Pty) Ltd, where I worked alongside experienced developers and gained valuable insight into a professional software development environment. I worked as a Student Supervisor at North-West University, assisting with marking tests, facilitating practical sessions, and tutoring students.

I enjoy learning new technologies, solving challenging problems and building software that makes a difference. I believe my technical skills, practical project experience and willingness to learn would allow me to contribute positively to Entelect's Graduate Programme.

Thank you for considering my application. I would welcome the opportunity to discuss my application further.

Yours sincerely,

Delius Schoeman

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SOFTWARE DEVELOPER | FULL-STACK WEB DEVELOPER | AI & MACHINE LEARNING

PROFESSIONAL SUMMARY

Information Technology graduate currently completing an Honours degree in Information Technology at North-West University, expected in November 2026. Passionate about software development, full-stack web development, artificial intelligence and machine learning. Experienced in developing software solutions through academic and personal projects, including full-stack web applications, third-party API integrations and machine learning research. Strong problem-solving and analytical skills with a commitment to developing reliable, scalable and user-focused software solutions.

EDUCATION

Bachelor of Information Technology Honours, North-West University

Expected Completion: November 2026

Research Topic: Multimodal Machine Learning Framework for the Early Detection and Prediction of Foot-and-Mouth Disease Outbreaks in South Africa.

Bachelor of Information Technology, North-West University

Graduated: 2025

ACHIEVEMENTS

- Top Performing Student – Achieved the highest final mark (93%) in an Information Technology module (CMPG 122 – UI Programming) at North-West University.
- Highest Achievement in Information Technology – Grade 12 & 11.

PROJECTS

Multimodal Machine Learning Framework for Foot-and-Mouth Disease Detection (Honours Research)

- Python, PyTorch, OpenCV, Machine Learning.
- Developing a multimodal ML framework integrating image, textual and structured environmental data.

Campus Marketplace Web Application

- Firebase, React, Node.js, CSS, Paystack API.
- Built a full-stack marketplace with authentication, payments and a relational database.
- Group Project

Marketplace Website

- PHP, MySQL, HTML, CSS, JavaScript, Google Maps API, Paystack API, The Courier Guy API.
- Developed a full-stack marketplace with authentication, payment processing, location services and delivery integration.
- Solo project

Personal Portfolio Website

- HTML, CSS, JavaScript.
- Responsive portfolio showcasing technical skills and projects.
- Live: ystervarkies.github.io/Delius-Schoeman-Portfolio

Morabaraba - AI Game Agent

- Python, Tkinter, Reinforcement Learning (Q-Learning).
- Built a self-learning Morabaraba game-playing agent for a university AI module, using tabular Q-learning with epsilon-greedy exploration and self-play training, including formal TEP (Task, Environment, Performance) framework design and experimental evaluation.

DETAILS

📞 079 218 3025

✉️ deliusschoeman.ds@gmail.com

LINKS

[Portfolio](#) [Website](#)

[GitHub](#)

[LinkedIn](#)

TECHNICAL SKILLS

- Python
- C#
- C++
- PHP
- JavaScript
- SQL
- HTML5
- CSS3
- React
- Next.js
- ASP.NET
- Bootstrap
- MySQL
- Firebase
- Machine Learning
- Computer Vision
- PyTorch
- Tensorflow
- OpenCV
- API
- Git
- GitHub
- Visual Studio
- Visual Studio Code
- XAMPP
- MATLAB

- Developed a Python/Tkinter GUI with JSON-based model export, and produced a technical report analysing agent performance.

EXPERIENCE

IT Vacation Student, Iclix (Pty) Ltd - July 2024

- Assisted with software development and IT-related tasks.
- Worked alongside experienced developers and gained exposure to software development workflows.

Construction Labourer (Vacation Employment) - recurring seasonal work: December 2022, December 2023, December 2024, December 2025, July 2026

- Worked during July and December university vacations.
- Developed teamwork, communication, reliability and time management skills.

Structured Programming (CMPG 121) Student Assistant – July 2026 until October 2026

- Assisted students with fundamental programming concepts in CMPG121 (Structured Programming), providing guidance during practical sessions, answering programming-related questions, and supporting the development of problem-solving and coding skills using C++.

Apps and Advanced User Interface Programming (CMPG212) Student Assistant – February 2026 until May 2026

- Assisted students with practical programming concepts in CMPG212 (Apps and Advanced User Interface Programming), providing guidance during practical sessions, answering technical questions, and supporting the development of C# and .NET applications.

User Interface Programming (CMPG122) Student Assistant – May 2024 until October 2024

- Assisted students with practical programming concepts in CMPG122 (Apps and Advanced User Interface Programming), providing guidance during practical sessions, answering technical questions, and supporting the development of C# and .NET applications.

PROFESSIONAL SKILLS

Software Development • Full-Stack Web Development • Machine Learning • Problem Solving • Analytical Thinking • Communication • Teamwork • Adaptability • Critical Thinking • Time Management

REFERENCES

Professor Dimitri Keykaan

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